Hybrid CNN-ViT for Edge Devices: Accuracy vs. Deployability Trade-off in Plant Disease Classification

I. Literature Review

Table I surveys fifteen studies, twelve of which were published in 2021 or later (nine of them in 2023–2025), spanning the foundational CNN and ViT architectures used in this project (MobileNetV2, MobileViT / MobileViTv2, ShuffleNet V2, and the original ViT), the PlantVillage baseline literature, and recent hybrid CNN-ViT and lightweight-ViT approaches applied specifically to plant disease classification. For each study the table reports the method, dataset, evaluation metrics used, the headline performance score(s), and the key limitation(s) noted by the authors or evident from the reported evaluation.

II. Critical Discussion of the Reviewed Literature

A. Foundational efficiency-oriented CNN architectures

MobileNetV2 [2] and ShuffleNet V2 [3] established the design principles this field still relies on: depthwise-separable and inverted-residual convolutions to cut FLOPs, and, in Ma et al.'s case, the empirical finding that FLOPs and real device latency diverge once memory-access cost and platform characteristics are taken into account. Neither paper touches vision transformers or agricultural data, so their contribution to this project is methodological: it justifies benchmarking MobileNetV2 on measured inference latency rather than FLOPs alone.

B. Foundational transformer and hybrid mobile architectures

ViT [1] established that pure self-attention can match CNN accuracy, but only with large-scale pretraining or heavy data augmentation; MobileViT [4] and MobileViTv2 [5] are direct responses that graft self-attention onto a convolutional backbone specifically to make ViT-style global context viable at mobile parameter and latency budgets. Read together, these three papers trace a clear trajectory (from "transformers need huge pretraining data" to "transformers need a hybrid convolutional front end" to "transformers need a cheaper attention operator") that this project's choice of MobileViT sits at the end of.

C. PlantVillage-only CNN and hybrid results

Mohanty et al. [6], Karthikeyan et al.'s CNN-SEEIB [10], and Pal et al.'s Mob-Res [11] all report accuracy above 99% on PlantVillage, a result that has been consistent across nearly a decade of publications and suggests the dataset itself, rather than the architecture, is now close to the accuracy ceiling. None of the three benchmarks a hybrid ViT baseline directly, and only CNN-SEEIB reports measured inference latency, so head-to-head deployability comparisons against a ViT-based model must be reconstructed rather than read directly off these papers.

D. Hybrid CNN-ViT and lightweight-ViT models built for plant disease

This is the cluster most directly relevant to the project's research question: Thakur et al. [7], PMVT [8], MobilePlantViT [9], Multi-ViT [13], PLA-ViT [14], and Aboelenin et al.'s hybrid framework [15] all claim accuracy parity with, or improvement over, CNN baselines, and several (PMVT, MobilePlantViT) explicitly target mobile deployment through parameter reduction. However, deployability reporting across the six is inconsistent: parameter count is reported more often than FLOPs, and FLOPs more often than measured latency; only PMVT and PLA-ViT discuss inference time explicitly, and neither publishes it alongside a like-for-like MobileNetV2 comparison. This inconsistency is the central evidence for the research gap this project addresses.

E. Deployability-focused benchmarking on field-realistic data

Kumar et al. [12] is the only reviewed paper that benchmarks several lightweight CNNs directly against each other on a combined, field-realistic dataset (PlantDoc + PlantVillage + PlantWild) rather than lab-only PlantVillage imagery. Its substantially lower accuracy ceiling (94.7%, versus the $>$99% figures reported elsewhere in Table I) is an important caution for interpreting the PlantVillage-only results throughout the rest of the literature. This project scopes its own evaluation to PlantVillage alone, but explicitly notes this caution as a limitation to be addressed in future work rather than within the present study.

TABLE I. LITERATURE SURVEY TABLE

Reference	Year	Method	Dataset	Evaluation Metrics	Performance Scores	Key Limitations
Dosovitskiy et al. [1]	2021	ViT: pure transformer encoder over 16×16 image patches; no convolutional inductive bias; large-scale pretraining then fine-tuning	ImageNet-21k / JFT-300M, CIFAR-100, VTAB	Top-1 Accuracy	ViT-H/14 ∼88.55% top-1 after JFT-300M pretraining; trained-from-scratch ViT-S/16 only 78.1% (below CNN baselines)	Needs very large-scale pretraining data to be competitive; underperforms CNNs when trained from scratch on smaller datasets; no built-in spatial inductive bias
Sandler et al. [2]	2018	MobileNetV2: inverted-residual blocks with linear bottlenecks and depthwise-separable convolutions	ImageNet, COCO, PASCAL VOC	Top-1 Accuracy, Params, MAdds (FLOPs)	72.0% top-1 on ImageNet with ∼3.4M parameters	Purely convolutional / local receptive field only; no global context modeling; accuracy plateaus below larger or hybrid models
Ma et al. [3]	2018	ShuffleNet V2: practical CNN design guidelines derived from measured on-device speed, not FLOPs alone	ImageNet	Top-1 Accuracy, Latency (ms), FLOPs	State-of-the-art speed-accuracy trade-off among CNNs at matched FLOPs on GPU and ARM platforms	Shows FLOPs is a poor latency proxy, but does not evaluate transformer or hybrid architectures at all; guidelines are platform-dependent
Mehta & Rastegari [4]	2022	MobileViT: hybrid CNN-ViT block that treats the transformer as a convolution; light-weight, mobile-friendly	ImageNet-1k, MS-COCO, PASCAL VOC	Top-1 Accuracy, Params, Latency	MobileViT-S: 78.4% top-1 with ∼5.6–6M params, 3.2 pp above MobileNetV3 and 6.2 pp above DeiT at similar size	Despite low FLOPs, multi-headed self-attention still gives higher latency than pure CNNs of comparable parameter count
Mehta & Rastegari [5]	2022	MobileViTv2: separable self-attention (linear complexity) replacing multi-head attention to cut latency	ImageNet-1k, MS-COCO	Top-1 Accuracy, Latency, Params	Runs ∼3.2× faster than MobileViT on mobile hardware at comparable or better accuracy	Separable attention trades some of full multi-head attention's representational flexibility for the speed gain
Mohanty et al. [6]	2016	Deep CNN (AlexNet / GoogLeNet, transfer learning) baseline; established PlantVillage as a benchmark	PlantVillage (54,306 images; 14 crops, 26 diseases)	Accuracy	99.35% accuracy on held-out PlantVillage test set	Images captured under controlled lab conditions with uniform backgrounds; accuracy drops substantially on field-collected images
Thakur et al. [7]	2023	Hybrid CNN-ViT model fusing convolutional local features with transformer self-attention; Grad-CAM/LIME explainability	5 public plant disease datasets	Accuracy, Precision, Recall, F1	Reported to outperform 9 state-of-the-art transfer-learning models across all 5 datasets (per-dataset scores, not a single pooled figure)	No FLOPs / latency / model-size benchmarking against a lightweight CNN baseline; explainability analysis is qualitative
Li et al. (PMVT) [8]	2023	Plant-based MobileViT: inverted-residual 7×7 conv block plus CBAM attention fused into the ViT encoder	PlantVillage, rice leaf, multiple field datasets	Accuracy, Params, Model size	93.1% accuracy on rice dataset, 3.4 pp above MobileNetV3 at similar size	Authors note lightweight ViTs remain slower than comparable lightweight CNNs at inference despite fewer parameters
Tonmoy et al. [9]	2025	MobilePlantViT: ultra-lightweight hybrid ViT (0.69M params) for generalized plant disease classification	Multiple plant disease datasets (varying scale)	Test Accuracy, Parameter count	0.69M parameters; 80–99%+ test accuracy depending on dataset difficulty	Wide accuracy spread (as low as 80%) shows generalization is strongly dataset-dependent; latency/FLOPs not reported
Karthikeyan et al. [10]	2025	CNN-SEEIB: Squeeze-and-Excitation-enabled identity blocks embedded in a lightweight CNN backbone	PlantVillage (38 classes, 54,305 images)	Accuracy, Precision, Recall, F1, Inference time	99.79% accuracy, F1 = 0.9971, 64 ms/image inference; 97.77% on an independent potato leaf dataset	Evaluated primarily on lab-condition PlantVillage images; no direct comparison against a ViT-based or hybrid baseline
Pal et al. (Mob-Res) [11]	2025	Mob-Res: MobileNetV2 backbone fused with parallel residual blocks; Grad-CAM-based interpretability	PlantVillage; Plant Disease Expert (58 classes)	Accuracy, Cross-domain validation	99.47% accuracy on PlantVillage; 97.73% on the larger Plant Disease Expert dataset; 3.51M parameters	Still ∼5× larger than the smallest hybrid ViT surveyed (MobilePlantViT, 0.69M); no measured latency reported
Kumar et al. [12]	2025	Head-to-head benchmark of MobileNetV2/V3 and EfficientNet-B0/B1 on combined field-realistic data	PlantDoc + PlantVillage + PlantWild (101 classes, 33 crops)	Accuracy, Params, Inference speed	Best model (EfficientNet-B1): 94.7% accuracy	Accuracy far below lab-only benchmarks, indicating PlantVillage-only results elsewhere likely overstate real-world performance; no ViT/hybrid model included
Baek [13]	2025	Multi-ViT: attention-score-based ensemble of multiple ViT branches for multi-leaf disease diagnosis	Apple, grape and tomato leaf disease datasets	Accuracy, F1	>99% accuracy with significant F1 gains over ResNet, VGG and MobileNet baselines	Requires multiple leaf images and multiple ViT branches per prediction, increasing inference cost and memory versus a single-image backbone
Murugavalli & Gopi (PLA-ViT) [14]	2025	PLA-ViT: transfer-learned ViT with bilateral filtering, data augmentation and adaptive learning-rate scheduling	Multiple plant leaf disease datasets	Accuracy, Inference time, Computational complexity	Reported to outperform baseline CNN/GAN models on accuracy, localization, inference time and complexity (exact figures not disclosed in the openly available abstract)	Comparative performance numbers are not fully reported in the open-access summary, limiting direct benchmarking against this project's models
Aboelenin et al. [15]	2025	Hybrid framework: CNN ensemble (VGG16, Inception-V3, DenseNet) for global features + ViT for local features	Apple and corn leaf disease datasets (4 classes each)	Accuracy	99.24% accuracy (apple), 98% accuracy (corn)	CNN ensemble + ViT fusion adds substantial architectural complexity and parameters; no inference-latency or FLOPs comparison reported

III. Introduction

A. Importance

Plant diseases destroy a substantial share of global crop yield every year, threatening food security and rural incomes, and their impact is heaviest precisely where diagnostic expertise is scarcest: smallholder farms in low- and middle-income regions [6]. Because disease symptoms often spread quickly across a field, the value of a diagnosis falls sharply with delay, which is why early, on-the-spot identification is treated as a priority in precision-agriculture research [9].

B. Background

Deep learning has been the dominant tool for automating this diagnosis since Mohanty et al. [6] showed that a convolutional neural network (CNN) fine-tuned on the PlantVillage dataset could identify 26 diseases across 14 crop species with 99.35% accuracy. That result opened two, largely separate research threads. The first pursued raw accuracy, first with deeper CNNs and later with Vision Transformers (ViTs) [1], which replace convolution with global self-attention and have matched or exceeded CNN accuracy on large-scale benchmarks, though at higher parameter counts and data requirements. The second thread optimized CNNs for the resource-constrained hardware on which a working diagnostic tool must actually run (smartphones and embedded field sensors), producing architectures such as MobileNetV2 [2] and ShuffleNet V2 [3], whose designers explicitly showed that FLOPs alone are a poor proxy for real device latency. Hybrid CNN-ViT architectures, most notably MobileViT [4] and its faster successor MobileViTv2 [5], were proposed to bridge these threads by combining convolutional local feature extraction with transformer-based global context at mobile-friendly parameter budgets. This hybrid idea has since been adapted directly to plant disease classification, yielding models such as PMVT [8] and MobilePlantViT [9], which report strong accuracy at very low parameter counts.

C. Problem Statement

Despite this progress, it remains unclear whether the accuracy gained by adopting a hybrid CNN-ViT architecture for plant disease classification is large enough, in a given deployment setting, to justify the model size and inference-latency cost it adds relative to a purely convolutional lightweight baseline. Reported results across the literature are difficult to compare directly because studies vary in dataset difficulty (controlled PlantVillage imagery versus field-collected "in-the-wild" images [12]), in which deployability metrics they report, and in whether latency is measured on real hardware or approximated through FLOPs [3]. Stated precisely, the problem this project addresses is the following: given the same plant disease classification task and dataset (PlantVillage), it is unclear whether a hybrid CNN-ViT model (MobileViT) provides a deployability-adjusted advantage over a pure lightweight CNN (MobileNetV2), that is, whether any accuracy improvement it offers is large enough to justify the increased model size and inference latency it adds for edge deployment.

D. Research Gap

Synthesizing across the fifteen studies reviewed, the recurring gap is that no reviewed paper benchmarks a hybrid CNN-ViT model and a pure lightweight CNN on the same plant disease dataset, under identical training conditions, across all four deployability axes used in general-purpose architecture research: accuracy, model size, FLOPs, and measured inference latency. Plant-disease-specific papers report accuracy, and sometimes parameter count, but rarely benchmark inference latency, model size, and FLOPs together against a matched lightweight CNN baseline trained under identical conditions [6, 7, 8, 9, 10, 11, 13, 14, 15]. Where deployability is benchmarked rigorously across accuracy, latency, parameters and FLOPs together, it is almost always on generic ImageNet-scale vision tasks rather than on plant disease data [1, 2, 3, 4, 5]. This leaves a gap between architecture-design papers, which quantify the accuracy-latency trade-off in general, and plant-disease papers, which mostly quantify accuracy alone.

E. Evidence

The trade-off this project investigates is visible even across the papers surveyed here. MobileViT reports a 3.2-percentage-point accuracy gain over MobileNetV3 at a comparable parameter count [4], but its successor, MobileViTv2, exists specifically because the original MobileViT's self-attention operations made it slower on-device than its FLOP count implied, a redesign that bought a $3.2\times$ latency reduction [5]. On plant disease data specifically, results diverge sharply by dataset difficulty: models evaluated on the controlled PlantVillage images report accuracies above 99% [6, 10, 11], while the same class of lightweight architectures evaluated on combined field datasets (PlantDoc, PlantVillage, PlantWild) top out closer to 94.7% [12], and the smallest hybrid ViT surveyed, Mobile-PlantViT (0.69M parameters), reports accuracies ranging from 80% to over 99% depending on the dataset [9], evidence that parameter-count reduction alone does not guarantee that accuracy holds.

F. Aims

This project aims to determine, for plant disease classification specifically, whether a hybrid CNN-ViT architecture (MobileViT) offers a deployability-adjusted advantage over a pure lightweight CNN (MobileNetV2) when both are fine-tuned under identical conditions on the same dataset.

G. Objectives

To this end, the project will (i) fine-tune MobileNetV2 and MobileViT on the PlantVillage dataset under matched training conditions; (ii) benchmark both models on accuracy, F1-score, model size, FLOPs, and inference latency; (iii) quantify and visualize the resulting accuracy-vs-deployability trade-off; and (iv) provide an evidence-based recommendation on which architecture is preferable under different edge-deployment constraints.

IV. Methodology and Experimental Setup

A. Proposed Methodology

This study will use a controlled comparative methodology to evaluate two lightweight deep learning models for plant disease classification: MobileNetV2 and MobileViT-XS. MobileNetV2 will be used as the lightweight CNN baseline, while MobileViT-XS will represent the hybrid CNN and Vision Transformer approach.

The main purpose of the experiment is to determine whether the use of transformer-based global feature learning in MobileViT provides a useful improvement in classification performance when its deployment cost is also considered. The study will therefore compare the models not only on predictive performance, but also on computational and deployment-related measures.

Both models will be evaluated on the same PlantVillage dataset. The same training, validation, and test split will be used for both models. The data preparation procedure, augmentation policy, training budget, model selection procedure, hardware environment, and evaluation process will also be kept the same wherever possible. This is necessary so that differences in the final results can be linked mainly to the model architectures rather than to different experimental conditions.

Both models will use ImageNet-pretrained weights and will then be fine-tuned for the 38 PlantVillage classes. Transfer learning is used because both architectures can benefit from visual features learned from a large general-purpose image dataset before being adapted to plant disease images. It also reduces the amount of training required compared with learning all model parameters from random initialization.

The proposed workflow is:

1) Load the original colour images from the PlantVillage dataset.
2) Create stratified training, validation, and test sets.
3) Apply the selected preprocessing steps and training-time data augmentation.
4) Load ImageNet-pretrained MobileNetV2 and MobileViT-XS models.
5) Replace their original classification layers with classifiers for the 38 PlantVillage classes.
6) Fine-tune both models using the same experimental protocol where technically appropriate.
7) Select the best model checkpoints using validation-set performance.
8) Evaluate both final models on the same unseen test set.
9) Measure classification performance using accuracy and F1-score.
10) Measure deployment-related properties using parameter count, model size, FLOPs, and inference latency.
11) Compare the results to study the trade-off between predictive performance and deployability.

The final objective is not only to identify which model achieves higher classification performance. The experiment will examine whether any performance difference is large enough to justify the computational cost of the model when used in an edge-device setting.

B. Model Architecture

Two architectures will be used in the study: MobileNetV2 and MobileViT-XS. They were selected because both target efficient visual recognition, but they use different methods for learning image features.

1) MobileNetV2: MobileNetV2 is used as the lightweight CNN baseline [2]. It was designed for mobile and resource-constrained environments. Its main building block is the inverted residual block with a linear bottleneck.

Instead of relying mainly on standard convolutions, MobileNetV2 uses depthwise separable convolutions to reduce computation. Within an inverted residual block, the input channels are first expanded, a depthwise convolution processes the spatial information, and a linear projection reduces the representation to a smaller number of channels. Residual connections are used when the input and output dimensions allow them.

These design choices make MobileNetV2 suitable as the baseline for this study because it represents a well-established approach to efficient CNN-based image classification. The standard MobileNetV2 model has approximately 3.5 million parameters, which is small compared with many conventional CNN architectures.

For this project, an ImageNet-pretrained MobileNetV2 model will be used. Its original ImageNet classification layer will be replaced with a new classification layer that produces 38 outputs, one for each PlantVillage class. The model will then be fine-tuned on the PlantVillage training set.

2) MobileViT-XS: MobileViT combines convolutional processing with transformer-based feature learning [4]. The CNN components learn local image patterns such as edges, shapes, textures, and disease spots. MobileViT blocks then use self-attention to model relationships between features from different spatial regions of the image.

A MobileViT block first uses convolutional layers to encode local spatial information. The resulting feature map is divided into small patches and passed through transformer layers so that information can be exchanged across different parts of the image. The transformed features are then returned to their spatial form and fused with convolutional features.

The MobileViT-XS variant will be used in this study. MobileViT-XS is selected instead of MobileViT-S because it provides a more suitable lightweight comparison with MobileNetV2. In the original MobileViT study, MobileViT-XS contains about 2.3 million parameters, while the standard MobileNetV2 contains about 3.5 million parameters. Using MobileViT-XS reduces the chance that the comparison is mainly determined by one model simply having a much larger parameter budget.

This choice is also useful for the deployability question. The original MobileViT experiments showed that MobileViT-XS can have fewer parameters than MobileNetV2 while still requiring more FLOPs and higher inference time. This means that parameter count alone is not enough to describe whether a model is suitable for edge deployment. Measuring actual inference latency is therefore an important part of this project.

As with MobileNetV2, ImageNet-pretrained MobileViT-XS weights will be used. The original classification head will be replaced with a new 38-class output layer before fine-tuning on PlantVillage.

3) Basis of the Architecture Comparison: The two models provide a direct comparison between two different lightweight design approaches.

TABLE II. MODEL ROLES IN THE COMPARATIVE STUDY

Model	**Main Approach**	**Main Role in the Study**
MobileNetV2	Lightweight convolutional neural network	CNN baseline
MobileViT-XS	Hybrid convolutional and transformer network	Hybrid comparison model

MobileNetV2 mainly uses convolutional operations and is optimized for efficient local feature extraction. MobileViT-XS keeps convolutional processing but adds transformer-based global feature learning. Comparing the two under the same experimental setup will allow the study to test whether the added global feature modelling leads to a useful accuracy improvement and how that improvement affects computational cost and inference speed.

The internal architectures of the pretrained models will not be redesigned. Only the final classification heads will be adapted for the 38 PlantVillage classes. This keeps the study focused on comparing the established architectures rather than introducing additional architectural changes that could make the results harder to interpret.

C. Dataset and Preprocessing

This study uses the PlantVillage dataset in its original colour format, consisting of 54,306 images spanning 38 classes across 14 crop species. The dataset will be split into training, validation, and test sets using an 80/10/10 ratio. The split will be stratified by class label to ensure that the proportion of each disease class remains consistent across all three sets, preventing underrepresentation of minority classes during evaluation.

Input images will be resized according to each model's native pretrained resolution: 224×224 for MobileNetV2 and 256×256 for MobileViT-XS, matching the resolution each architecture was originally pretrained on. This choice preserves the integrity of each model's pretrained feature representations, avoiding the risk of artificially disadvantaging either model by forcing it outside its native input size.

Pixel values will be normalized using the standard ImageNet mean and standard deviation values ([0.485, 0.456, 0.406] and [0.229, 0.224, 0.225] respectively, per RGB channel), consistent with the normalization statistics used during the original pretraining of both models.

Training-time data augmentation will include random horizontal flipping, small random rotations ($\pm 15^{\circ}$), and mild colour jitter (brightness and contrast). These augmentations are intended to improve generalization without distorting disease-relevant visual features such as lesion shape, colour, or spotting pattern, which are diagnostically meaningful and should not be aggressively altered. Validation and test images will not be augmented, and will only undergo resizing and normalization, to ensure evaluation reflects realistic, unmodified inputs.

The same augmentation policy will be applied identically to both models, with only the input resolution differing between them, consistent with the controlled comparative methodology described in Section 3.1.

The experiments will be implemented in PyTorch, which natively supports both architectures with verified pretrained checkpoints, reducing the risk of implementation inconsistency between the two models.

D. Training Configuration

Both models will be fine-tuned under matched training conditions to isolate architecture as the primary source of any performance difference.

Optimizer: AdamW will be used for both models. AdamW is standard practice for fine-tuning pretrained vision models, as its decoupled weight decay improves generalization compared to standard Adam.

Learning rate: An initial learning rate of 1e-4 will be used for both models, a commonly used starting point for fine-tuning pretrained convolutional and transformer-based vision models on moderately sized datasets.

Learning rate scheduler: A cosine annealing schedule will be used to gradually reduce the learning rate over training, which helps stabilize convergence in later epochs.

Batch size: A batch size of 32 will be used for both models, selected to balance training stability with the memory constraints of the Google Colab environment used for training.

Epochs: Both models will be fine-tuned for a maximum of 20 epochs, with the best checkpoint selected based on validation performance rather than final-epoch performance.

Loss function: Categorical cross-entropy loss will be used, as this is the standard loss function for multi-class classification tasks with mutually exclusive class labels.

Early stopping: Training will be stopped early if validation loss does not improve for 5 consecutive epochs, to prevent overfitting and reduce unnecessary computation.

Regularization: Weight decay of 1e-4 will be applied via the AdamW optimizer. No additional dropout layers will be introduced beyond those already present in the pretrained architectures, to avoid altering the original model designs.

Random seed: A fixed random seed (42) will be used across data splitting, model initialization, and training for both models, to support reproducibility.

Hardware: Both models will be trained on the same Google Colab GPU runtime, ensuring that any timing or performance differences are not attributable to differing hardware environments.

This configuration will be applied identically to both MobileNetV2 and MobileViT-XS wherever architecturally appropriate, in line with the controlled comparison stated under methodology.

E. Evaluation Metrics

Model performance will be evaluated using a combination of predictive accuracy metrics and deployability metrics, reflecting the project's dual focus on classification performance and edge-device suitability.

Accuracy will be calculated as the proportion of correctly classified test images out of the total number of test images. Accuracy is reported because it is the most widely used metric in the reviewed literature, allowing direct comparability with prior PlantVillage studies.

F1-score will be reported using the macro-averaged variant, calculated independently for each of the 38 classes and then averaged without weighting by class frequency. Macro-averaging is used rather than weighted averaging because it treats all disease classes equally regardless of how many images represent them, which better reflects real-world diagnostic importance, since a rare disease class is not less important to correctly identify than a common one.

Model size will be reported in two forms: the total number of trainable parameters, and the on-disk size of the saved model file in megabytes. Parameter count allows direct comparison with figures reported in the original MobileNetV2 and MobileViT papers, while file size reflects practical storage requirements on a mobile device.

FLOPs (floating-point operations) will be calculated for a single forward pass at each model's respective input resolution, using a standard model-profiling library. FLOPs provide a hardware-independent estimate of computational cost, allowing comparison with other studies that do not report measured latency.

Inference latency will be measured as the average wall-clock time required to process a single image, timed on the same hardware used for training. Latency will be measured over 100 repeated inference runs per model, with the first 10 runs excluded as warm-up to avoid including one-time initialization overhead. The same batch size and hardware will be used for both models to ensure a fair comparison.

Latency is included specifically because, as established in the literature review, FLOPs and parameter count are not reliable proxies for real-world inference speed; measuring latency directly is necessary to assess true edge-deployment suitability.

Together, these five metrics allow the study to evaluate not only which model classifies plant diseases more accurately, but whether any accuracy advantage is large enough to justify its cost in model size, computational complexity, and real-world inference speed, directly addressing the research question stated.

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